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Primary Noncontrast Magnetic Resonance Imaging for Prostate Cancer Screening: A Randomized Clinical Trial (PROSA)

Emanuele Messina [a], *Antonella Borrelli* [a], *Alessandro Sciarra* [b], *Ludovica Laschena* [a], *Debora Antonini* [c], *David Shaholli* [c], *Fabio M. Magliocca* [a], *Simone Novelli* [d,e], *Daniele Santini* [f], *Giuseppe La Torre* [c], *Valeria Panebianco* [a,*]

[a] *Department of Radiological Sciences, Oncology and Pathology, Policlinico Umberto I, Sapienza University, Rome, Italy;* [b] *Department of Maternal-Infant and Urological Sciences, Policlinico Umberto I, Sapienza University, Rome, Italy;* [c] *Department of Public Health and Infectious Diseases, Policlinico Umberto I, Sapienza University, Rome, Italy;* [d] *Department of Mechanical and Aerospace Engineering, Sapienza University, Rome, Italy;* [e] *Institute for Liver and Digestive Health, Royal Free Campus, University College London, London, UK;* [f] *Medical Oncology Unit, Policlinico Umberto I, Sapienza University of Rome, Rome, Italy*

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Abstract

Background and objective: Prostate-specific antigen (PSA)-based screening for prostate cancer (PCa) has limited accuracy, and it is linked to overdiagnosis. The PROSA trial aimed to evaluate whether a contrast-free biparametric magnetic resonance imaging (bpMRI)-first screening strategy improves the detection of clinically significant PCa (csPCa) as the primary outcome. The secondary outcomes included overall PCa detection, benefit-harm metrics, and cost effectiveness from a health care payer perspective.
Methods: This single-center, randomized controlled trial enrolled 816 asymptomatic men aged 49–69 yr (≥40 yr with a PCa family history). Participants were randomized into two arms: arm A underwent bpMRI regardless of the PSA levels; arm B received bpMRI only if PSA ≥3 ng/ml (or 2.5 ng/ml with a family history). Men with Prostate Imaging Reporting and Data System score ≥3 were directed to a targeted biopsy. Imaging and pathology assessors were blinded; csPCa is defined as International Society of Urological Pathology grade group ≥2. The primary outcomes included csPCa detection, benefit-harm metrics, and cost effectiveness from a health care payer perspective.
Key findings and limitations: Among 759 randomized men, biopsy and csPCa detection rates were higher in arm A (10.8% and 4.6%, respectively) than in arm B (5.2% and 1.8%, respectively), with a relative risk of 2.6 (95% confidence interval 1.1–6.1; $p = 0.05$) for the csPCa detection rate. Benefit-harm metrics favored the MRI-first strategy, showing higher grade selectivity (1.89 vs 1.75), biopsy efficiency (0.74 vs 0.54), and biopsy avoidance (23.1 vs 11.9). No serious adverse event was recorded. The MRI-first strategy yielded an incremental cost-effectiveness ratio of €2201.75 per csPCa case detected. Limitations include single-round design and short follow-up.

* Corresponding author. Department of Radiological Sciences, Oncology and Pathology, Sapienza/Policlinico Umberto I, Viale Regina Elena 324, 00161 Rome, Italy. Tel. +39 064 997 5463; Fax: +39 064 997 8509.
E-mail address: valeria.panebianco@uniroma1.it (V. Panebianco).

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Conclusions and clinical implications: In this randomized screening trial, a contrast-free MRI-first pathway improved csPCa detection, enhanced benefit-harm metrics, and showed favorable cost effectiveness.

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ADVANCING PRACTICE

What does this study add?
The PROSA trial demonstrates that contrast-free biparametric magnetic resonance imaging (bpMRI) can serve as a clinically effective and economically sustainable alternative to prostate-specific antigen–based prostate cancer screening. This MRI-first approach increases the detection rate of clinically significant prostate cancer significantly, without increasing biopsy rates or overdiagnosis. The favorable incremental cost-effectiveness ratio and the number needed to screen support its scalability in population-level screening. These findings provide new evidence for incorporating bpMRI into early detection strategies within health technology assessment frameworks.

Clinical Relevance
This randomized trial demonstrates that a contrast-free MRI-first screening strategy significantly improves detection of clinically significant prostate cancer over the standard PSA-based pathway, identifying nearly half of aggressive tumours in men with normal PSA levels. By enhancing biopsy efficiency and grade characterization, the MRI-first strategy offers a more favourable benefit-to-harm profile than PSA-triggered imaging. Its robust cost-effectiveness further supports the incorporation of biparametric MRI into early detection algorithms, providing uro-oncologists with a pragmatic and scalable approach to refine population-based prostate cancer screening. Associate Editor: Gianluca Giannarini, MD.

Patient Summary
In this study, we compared the standard prostate-specific antigen (PSA)-based screening with magnetic resonance imaging (MRI)-based screening for prostate cancer in men without symptoms. We found that the use of contrast-free MRI led to earlier detection of potentially clinically relevant cancer, even in men with normal PSA levels, without leading to more unnecessary biopsies. This approach offers a safe and effective way to screen for prostate cancer.

1. Introduction

Prostate cancer (PCa) is the most frequently diagnosed cancer in men, with an incidence rate of 154.1 cases per 100 000 individuals in European Union member states. It has a mortality rate of 38.6 per 100 000 men, making it the third leading cause of cancer-related deaths [1]. The economic burden of PCa is substantial; in 2017, the estimated global health care cost reached 624 billion international dollars (95% confidence interval [CI]: 473–905) [2].

PCa ranges from indolent to aggressive forms, necessitating accurate risk stratification. While clinically insignificant PCa (ciPCa) often permits conservative management to minimize overtreatment and associated side effects, clinically significant PCa (csPCa) may require prompt detection and intervention to prevent progression and improve patient outcomes [3,4]. Ensuring timely diagnosis allows patients to benefit from localized treatments, with improved outcomes, fewer complications, and a lower incidence of advanced disease [5,6].

Various PCa screening strategies have been explored, with prostate-specific antigen (PSA) testing being the most common. However, PSA lacks specificity, leading to false positives, overdiagnosis of ciPCa cases, and unnecessary interventions, while some aggressive cases may be missed due to false negatives [7]. In this context, magnetic resonance imaging (MRI) with a contrast media injection (multiparametric MRI [mpMRI]) proved its precise diagnostic performance in PCa detection [8–12], and recently MRI without a contrast media injection (biparametric MRI [bpMRI]) has emerged as a promising tool for improving PCa early detection [13–17].

Several trials, such as IP1-PROSTAGRAM (2021), MVP (2022) [18], PROBASE (2023), Göteborg-2 (2022), Stockholm3 (2024) [19], and PROSCREEN (2024) [20], investigated the role of MRI in PCa screening, using diverse methodologies and objectives, showing that incorporation of MRI into PCa screening could enhance detection efficiency.

However, none of the already published studies applied a health technology assessment (HTA) analysis to evaluate not only the efficiency, but also the cost effectiveness of the experimental screening pathways.

The primary outcome of the PROSA trial was the detection rate of csPCa (International Society of Urological Pathology [ISUP] grade group [GG] $\geq$2). The secondary outcomes included the overall PCa detection rate (csPCa and ciPCa), benefit-harm, as well as the cost effectiveness of

the MRI-first screening strategy from a health care payer perspective.

2. Patients and methods

2.1. Study design

PROSA is a prospective, randomized, interventional, single-center study, conducted at a university hospital in Italy [21]. It was formally approved by the institutional ethics committee (approval 09/09/2020, Ref. 5996). The study is registered and accessible at https://clinicaltrials.gov/ct2/show/NCT04803188. The study flowchart is illustrated in Supplementary Fig. 1, and it followed the Consolidated Standards of Reporting Trials (CONSORT) guidelines, as reported in Fig. 1.

2.2. Participants

A total of 816 men aged 49–69 yr (40 yr in case of a family history of PCa) have been enrolled. Written informed consent was obtained from all participants. Inclusion and exclusion criteria are listed in Table 1. All participants were volunteers, primarily employees of the hospital and the university where the trial was conducted. Targeted dissemination activities were organized to promote the study and facilitate participants' recruitment.

2.3. Study design, randomization, and follow-up

After collection of personal/clinical data during an initial medical screening, 1:1 randomization was performed, blindly directing participants into two different arms: in arm A, men underwent bpMRI regardless of their PSA values, and in arm B, men with increased PSA (>3 ng/ml, or 2.5 ng/ml if there is a positive family history for PCa) were referred to bpMRI, while those with normal PSA were not.

The randomization sequence was generated by an independent statistician using a computer-based random number generator, applying simple randomization without stratification or blocking. Allocation was implemented through a centralized, password-protected electronic data-

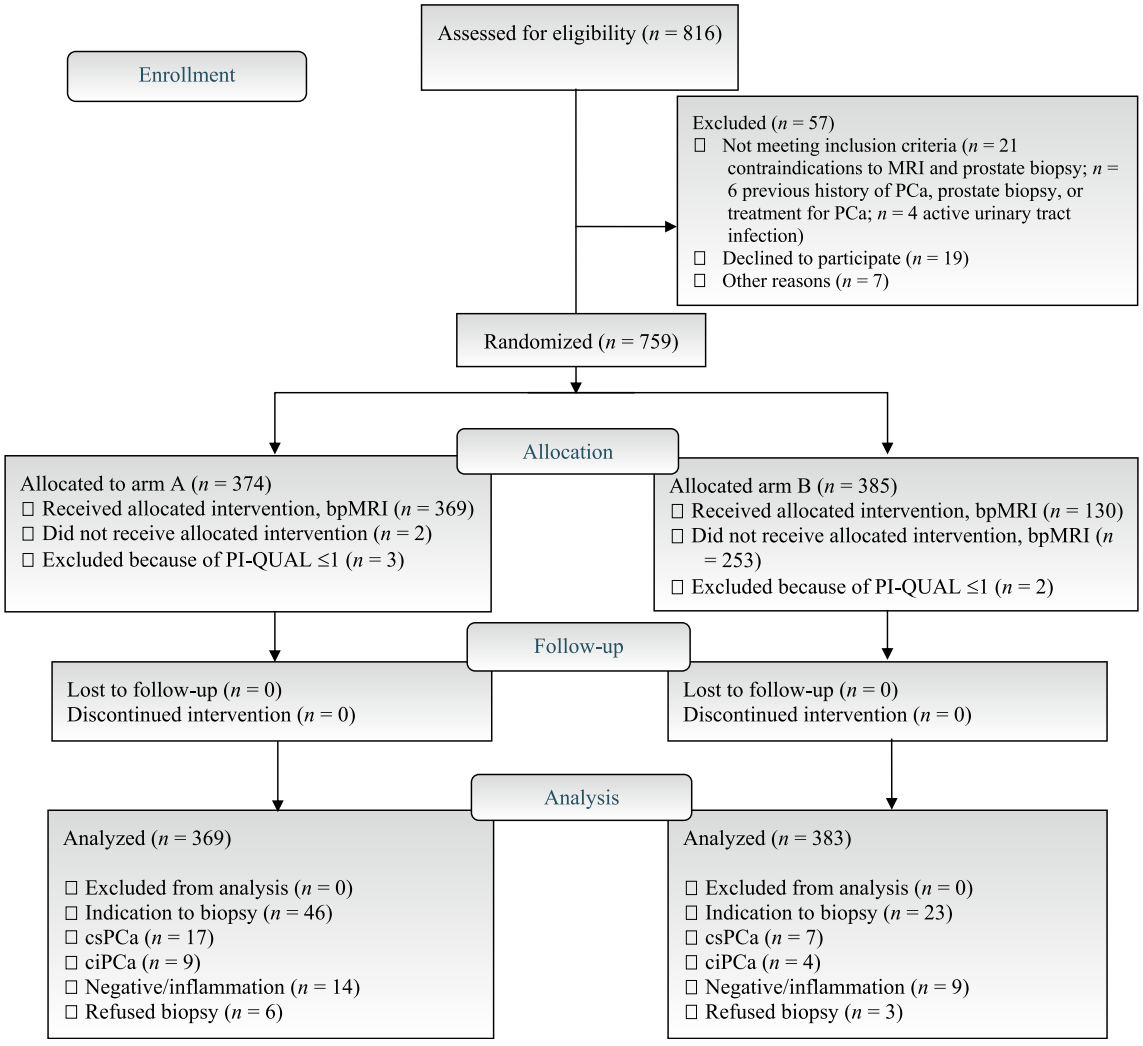


Fig. 1 – CONSORT flow diagram, showing the study flowchart. bpMRI = biparametric magnetic resonance imaging; ciPCa = clinically insignificant prostate cancer; CONSORT = Consolidated Standards of Reporting Trials; csPCa = clinically significant prostate cancer; MRI = magnetic resonance imaging; PCa = prostate cancer; PI-QUAL = Prostate Imaging Quality.

Table 1 – Exclusion and inclusion criteria of the study

Exclusion criteria	Inclusion criteria
1. Contraindications to MRI and prostate biopsy	1. Males aged between 49 and 69 yr at the time of enrollment, in absence of risk factors, and from the age of 40 yr for those with a family history of PCa
2. Previous history of PCa, prostate biopsy, or treatment for PCa	2. Life expectancy ≥10 yr
3. Active urinary tract infection and prostatitis (National Institutes of Health category I, II, and III)	3. Patient with full possession of one's faculties, and in general able to express the informed consent and to carry out all the examinations and procedures required by the study
4. Low-quality magnetic resonance images (PI-QUAL ≤1) [a]	

MRI = magnetic resonance imaging; PCa = prostate cancer; PI-QUAL = Prostate Imaging Quality.
[a] MR images of low quality have always been considered an exclusion criterion; after the publication of PI-QUAL v2, this standardized scoring system was applied retrospectively and prospectively to ensure consistent image quality assessment.

base that automatically assigned participants to one of the two arms upon confirmation of eligibility. The randomization list was inaccessible to the enrolling clinicians, ensuring allocation concealment until assignment.

Participant enrollment and randomization were handled by separate investigators to prevent a selection bias. Imaging and pathology assessors were blinded to group allocation and clinical data, while participants were unmasked due to the nature of the interventions. Masking integrity was maintained through strict separation of study roles.

Participants were monitored with serial PSA assessments throughout the study. In cases of significant PSA changes, mpMRI was performed as per the clinical guidelines. Those with initial negative bpMRI (Prostate Imaging Reporting and Data System [PI-RADS] score ≤2) also underwent at least one follow-up MRI scan after 12 mo, ensuring consistent follow-up.

Recruitment took place between 2020 and 2024; the last follow-up assessment was completed in 2025, at which point the study population was closed. Since the trial was designed as a single-round screening study, follow-up MRI data were collected but not included in the primary analysis.

All participants were monitored for adverse events following MRI and a biopsy. As bpMRI is a noncontrast, nonionizing technique, it is not associated with direct adverse effects. Potential contraindications to MRI (metallic implants and claustrophobia) were part of the exclusion criteria, so participants at risk were not enrolled. After a biopsy, men were observed for immediate complications and followed up for delayed events.

2.4. Biparametric MRI acquisition and reporting

All MRI examinations were conducted using one of two 3-Tesla scanners (GE Discovery 750 or Siemens MAGNETOM Vida) with a 32-channel surface pelvic coil, following PI-RADS v2.1 guidelines adapted for a biparametric protocol [22]. The imaging protocol included high-resolution T2-weighted imaging (T2WI) and diffusion-weighted imaging (b values of 100, 800, 1000, and 2000 s/mm^2) with apparent diffusion coefficient map reconstruction (from b values of 100, 800, and 1000 s/mm^2). To optimize image quality, patients followed a preparatory regimen including a low-fiber diet and evacuative enema [23]. Two experienced radiologists, blinded to the clinical data, independently assigned PI-RADS scores to visible lesions (ranging from 1 to 5); discrepancies were resolved by consensus or, if needed, by the more experienced reader. A PI-RADS score of ≥3 was used as a threshold to direct to a biopsy [24]. Considering magnetic resonance (MR) image quality to be of outmost importance [25,26], low quality of MR images has always been considered an exclusion criterion according to the study protocol. To standardize this evaluation, following the publication of Prostate Imaging Quality (PI-QUAL) version 2 [27], we retrospectively reassessed all MRI scans, including those initially excluded, confirming that previously excluded cases corresponded to PI-QUAL 1, while this standardized scoring system was prospectively applied to all subsequent examinations.

2.5. Prostate biopsy

Each patient who was recommended to undergo a prostate biopsy had an MR-directed biopsy (MRDB) using a targeted-only approach with the transrectal ultrasound–MRI fusion technique with a specialized system, without collecting systematic cores. The MRDB was performed by the same team of expert radiologists who acquired and reported the MR images.

A specialized pathologist with 15 yr of experience in genitourinary pathology analyzed and described the biopsy samples using consistent terminology; csPCa is defined as ISUP GG ≥2 [8].

2.6. Cost assessment and resource utilization

Cost data were collected and analyzed from a health care payer perspective, considering only direct medical costs typically reimbursed or incurred by the health care system. The sources used to inform the analysis are detailed in Supplementary Table 1.

This approach aligns with the standard practices in health economic evaluations focused on cost effectiveness from the payer's viewpoint.

2.7. Statistical and cost-effectiveness analyses

The initial sample size calculation was based on preliminary data from clinical cohorts, aiming to compare the proportion of participants with a positive screening result between

the two study arms. Specifically, we assumed that approximately 30% of men undergoing bpMRI (arm A) would show PI-RADS ≥3 lesions, compared with 40% of men screened by PSA (arm B) expected to have PSA ≥3 ng/ml. A two-sample comparison of proportions with a two-sided α of 0.05 and 80% power was applied.

Following the publication of population-based studies [18,28], the expected rates were updated to 20% for PI-RADS score ≥3 and 12% for PSA ≥3 ng/ml, resulting in a minimum required sample size of 328 participants per arm, confirming that the originally planned enrollment target was adequate.

The analysis was conducted on a per-participant basis, considering one index lesion per participant. In participants with multiple MRI-detected alterations, the index lesion was defined as the one with the highest PI-RADS score, or, if scores were equal, the lesion with the largest diameter. Baseline balance between the study arms was evaluated using standardized mean differences, with values >0.20 indicating potential imbalance. Between-group comparisons were expressed not only through p values, but also as effect size estimates. For binary outcomes, we calculated absolute risk differences (RDs) indicated as percentage points (pp) and relative risks (RRs), each with corresponding 95% confidence intervals (95% CIs). CIs for RDs were computed using Wald methods, and those for RRs were computed on the log scale. The number needed to treat (NNT) was derived as the reciprocal of the RD for positive effects, indicating how many bpMRI scans are needed to screen one additional case.

Continuous variables were compared using Mann-Whitney U tests, and differences in medians were reported with 95% CIs where applicable. Statistical significance was set at $p < 0.05$.

We performed a benefit-harm analysis as follows: grade selectivity (GG ≥2/GG1), biopsy efficiency (GG ≥2/[GG1 + benign]), and biopsy avoidance (avoided biopsies among MRI-negative men per benign biopsy among MRI-positive men).

Moreover, a comprehensive economic evaluation was conducted using an HTA framework, incorporating a microcosting approach based on the actual costs of procedures. The incremental cost-effectiveness ratio (ICER) is calculated as follows: Δcosts/Δeffectiveness = (CA − CB)/(EA − EB); CA represents the overall costs related to the pathway followed on arm A, while CB is the total costs linked to arm B; moreover, EA indicates the effectiveness of arm A, represented by csPCa detection rate, while EB represents the same data from arm B. We performed one-way deterministic sensitivity analyses on total costs and on effectiveness by varying each symmetrically by ±10%, ±20%, ±30%, ±40%, and ±50%, holding all other inputs constant. To estimate health outcomes for a cost-utility analysis, we converted the number of cancers detected into quality-adjusted life-years (QALYs), using an evidence-based estimate of 0.26 QALYs gained per cancer detected [29]. Application of this parameter to each screening strategy enabled calculation of the incremental cost-utility ratio (ICUR).

An exploratory analysis evaluating the alternative PI-RADS ≥4 biopsy threshold was conducted, to assess the potential impact of a higher threshold.

The statistical analysis was carried out using SPSS package, version 28.0 (IBM Corporation, Armonk, NY, USA).

An interim descriptive analysis of feasibility and preliminary outcomes was published previously [21], but no formal interim efficacy analyses or stopping rules were prespecified; the trial continued to completion as designed originally.

3. Results

3.1. Participant flow and baseline characteristics

A total of 816 men were assessed for eligibility. Of these men, 57 were excluded, as detailed in Fig. 1. The remaining 759 participants were assigned randomly to one of the two groups. In all, 369 participants were analyzed in arm A and 383 participants were analyzed in arm B (Fig. 1). No participants were lost to follow-up. The baseline characteristics are detailed in Table 2.

Only MRI classified with a PI-QUAL score of ≥2 were considered in the final cohort: 6.8% (34/499; 95% CI 4.9–9.4%) with a PI-QUAL score of 2 and 93.2% (465/499; 95% CI 90.6–95.1%) with a PI-QUAL score of 3.

3.2. Cancer detection rates

In arm A, 40 biopsies were performed (10.8% [40/369]; 95% CI 8.1–14.4%), with the diagnosis of 17 csPCa (4.6% [17/369]; 95% CI 2.9–7.3%) and nine ciPCa (2.4% [9/369]; 95% CI 1.3–4.5%) cases (Table 2).

In arm B, 20 biopsies were performed (5.2% [20/383]; 95% CI 3.3–8.0%), with the diagnosis of seven cases of csPCa (1.8% [7/383]; 95% CI 0.9–3.7%) and four cases of ciPCa (1.0% [4/383]; 95% CI 0.4–2.5%; Table 2).

A comparison of the histopathological outcomes between the arms are described in Table 2. More specifically, arm A showed higher detection rates of csPCa than arm B (4.6% vs 1.8%), corresponding to an RD of 2.8 (95 % CI 0.3–5.4) and an RR of 2.5 (95% CI 1.1–6.1). The rates of negative/inflammatory findings and low-grade (GG 1) cancers were comparable between the arms. The NNT to detect one additional case of csPCa was 36; on the contrary, if we consider the diagnosis of overall PCa (including both csPCa and ciPCa), the NNT is 24.

3.3. MRI-targeted biopsy results and benefit-harm analysis

The biopsy rates were 10.8% in arm A and 5.2% in arm B (RD = 5.6 pp, 95% CI 2.0–9.1; RR = 2.1, 95% CI 1.2–3.5). In arm A, eight men diagnosed with csPCa (47.1%; 95% CI 26.2–68.9%) presented normal PSA levels (Fig. 2).

A total of 60 MRDB procedures were conducted, with an average of 3.8 cores collected per MRI target for each patient. Clinical, radiological, and pathological characteristics of the men diagnosed with GG ≥2 PCa are listed in Supplementary Table 2.

Considering a benefit-harm analysis, grade selectivity was 1.89 in arm A (17/9) and 1.75 in arm B (7/4), biopsy

Table 2 – Baseline characteristics and radiological/clinical-pathological outcomes of the enrolled men, divided into the two arms of interest

Baseline characteristics	Arm A Median (quartiles) or no. (%)	Arm B Median (quartiles) or no. (%)	SMD/*p* value		
Age	60 (50–67)	59 (51–66)	0.12		
PCa family history			0.11		
No	294 (79.8)	322 (84.1)			
Yes	75 (20.2)	61 (15.9)			
BMI	25.76 (22.1–32.6)	25.67 (23.5–31.7)	0.19		
Chronic diseases			0.13		
No	262 (71.0)	249 (65.0)			
Yes	107 (29.0)	134 (35.0)			
Smoking history			0.08		
No	287 (77.8)	284 (74.2)			
Yes	82 (22.2)	99 (25.8)			
PSA	2.26 (0.7–3.2)	2.00 (0.6–2.6)	0.7		
Radiological/clinical-pathological outcomes	**Arm A Median (quartiles) or no. (%)**	**ARM B Median (quartiles) or no. (%)**	**Risk difference, pp (95% CI)**	**Relative risk (95% CI)**	***p* value**
PI-RADS score v2.1					
1	68 (18.4)	28 (21.5 [a])	−3.1 (−10.9 to 4.7)	0.85 (0.5–1.3)	0.52
2	255 (69.1)	79 (60.8 [a])	8.3 (−2.3 to 18.8)	1.14 (1.0–1.3)	0.10
3	25 (6.8)	10 (7.7 [a])	−0.9 (−5.7 to 3.9)	0.88 (0.4–1.9)	0.88
4	21 (5.7)	13 (10.0 [a])	−4.3 (−9.8 to 1.3)	0.57 (0.3–1.1)	0.14
5	0 (0)	0 (0 [a])	–	–	–
Histology and ISUP GG (GS)					
Negative/inflammation	14 (3.8)	9 (2.3)	1.5 (−0.6 to 3.6)	1.7 (0.7–3.8)	0.35
Prostate Cancer	26 (7.0)	11 (2.9)	4.2 (1.0–7.1)	2.5 (1.3–5.1)	0.01
GG1 (3 + 3)	9 (2.4)	4 (1.0)	1.4 (−0.3 to 3.1)	2.3 (0.7–7.5)	0.23
GG 2, 3, and 4 (3 + 4, 4 + 3, 4 + 4)	17 (4.6)	7 (1.8)	2.8 (0.3–5.4)	2.5 (1.1–6.1)	0.05

BMI = body mass index; CI = confidence interval; GG = grade group; GS = Gleason score; ISUP = International Society of Urological Pathology; MRI = magnetic resonance imaging; PCa = prostate cancer; PI-RADS = Prostate Imaging Reporting and Data System; pp = percentage points; PSA = prostate-specific antigen; SMD = standardized mean differences.

[a] Percentages calculated considering only those patients who underwent MRI.

efficiency was 0.74 in arm A (17/23) and 0.54 in arm B (7/13), and biopsy avoidance was 23.1 in arm A (323/14) and 11.9 in arm B (107/9). Moreover, the numbers needed to biopsy to detect one case of GG ≥2 cancer were 2.35 in arm A (40 biopsies/17 GG ≥2) and 2.86 in arm B (20/7).

Different PI-RADS distributions among csPCa, ciPCa, and negative/inflammation results are detailed in Supplementary Table 3. Among the 60 cases undergoing MRDB procedures, 26 (43.3%; 95% CI 30.8–55.9%) were found to be PI-RADS 3 (two csPCa, nine GG1, and 15 benign).

3.4. HTA-based cost-effectiveness analysis

All costs related to the different procedures involved in both arm A and arm B have been calculated with the cost-effectiveness approach, and these are detailed in Table 3. Considering the MRI-first strategy, ΔCost was €22 017.53 for ten additional csPCa cases, giving an ICER of €2201.75 per csPCa case. Variation of total costs in the range from ±10% to ±50% produced ICERs ranging from €1100.88 (−50%) to €3302.63 (+50%). Variation in effectiveness (ΔEffect) from ±10% to ±50% yielded ICERs from €1467.84 (+50%) to €4403.51 (−50%). The best case (−50% cost, +50% effect) gave an ICER of €733.92, while the worst case (+50% cost, −50% effect) gave an ICER of €6605.26 per csPCa case. Across these broad ranges, most scenarios remained clustered around ∼€2000–3000 per case of csPCa.

Applying the estimated value of 0.26 QALYs per cancer case detected, the MRI-first screening strategy generated a net gain of 2.6 QALYs compared with the PSA-first strategy. Given an incremental cost of €22 017.53, the resulting ICUR was €8468 per QALY gained.

3.5. Exploratory analysis with PI-RADS ≥4 biopsy threshold

Considering a threshold of PI-RADS ≥4, 26 of 60 MRDB procedures (43.3%, 95% CI 31.3–55.9%) would have been avoided; specifically, of these 26 cases, 19 (73.1%, 95% CI 52.2–88.4%) from arm A and seven (26.9%, 95% CI 11.6–47.8%) from arm B. Among the avoided biopsies, two of 24 csPCa cases (8.3%, 95% CI 1.0–27.0%), one in each arm, with nine of 13 GG1 (69.2%, 95% CI 38.6–90.9%) and 15/23 benign findings (65.2%, 95% CI 42.7–83.6%), would have been omitted. The positive predictive value for csPCa among biopsied men would increase from 40.0% (24/60) to 64.7% (22/34), while grade selectivity would rise from 1.85 (24/13) to 5.50 (22/4), biopsy efficiency from 0.67 (24/36) to 1.83 (22/12), and biopsy avoidance from 18.7 (430/23) to 58.1 (465/8). Using the observed average payer cost per biopsy of €471.70, the 26 avoided procedures correspond to €12 264.20 in savings, €8962.30 from arm A and €3301.90 from arm B, reducing total costs to €33 329.82 and €16 972.69, respectively. The resulting ΔCost of €16 357.13, combined with an unchanged ΔEffect of ten csPCa cases, yields an

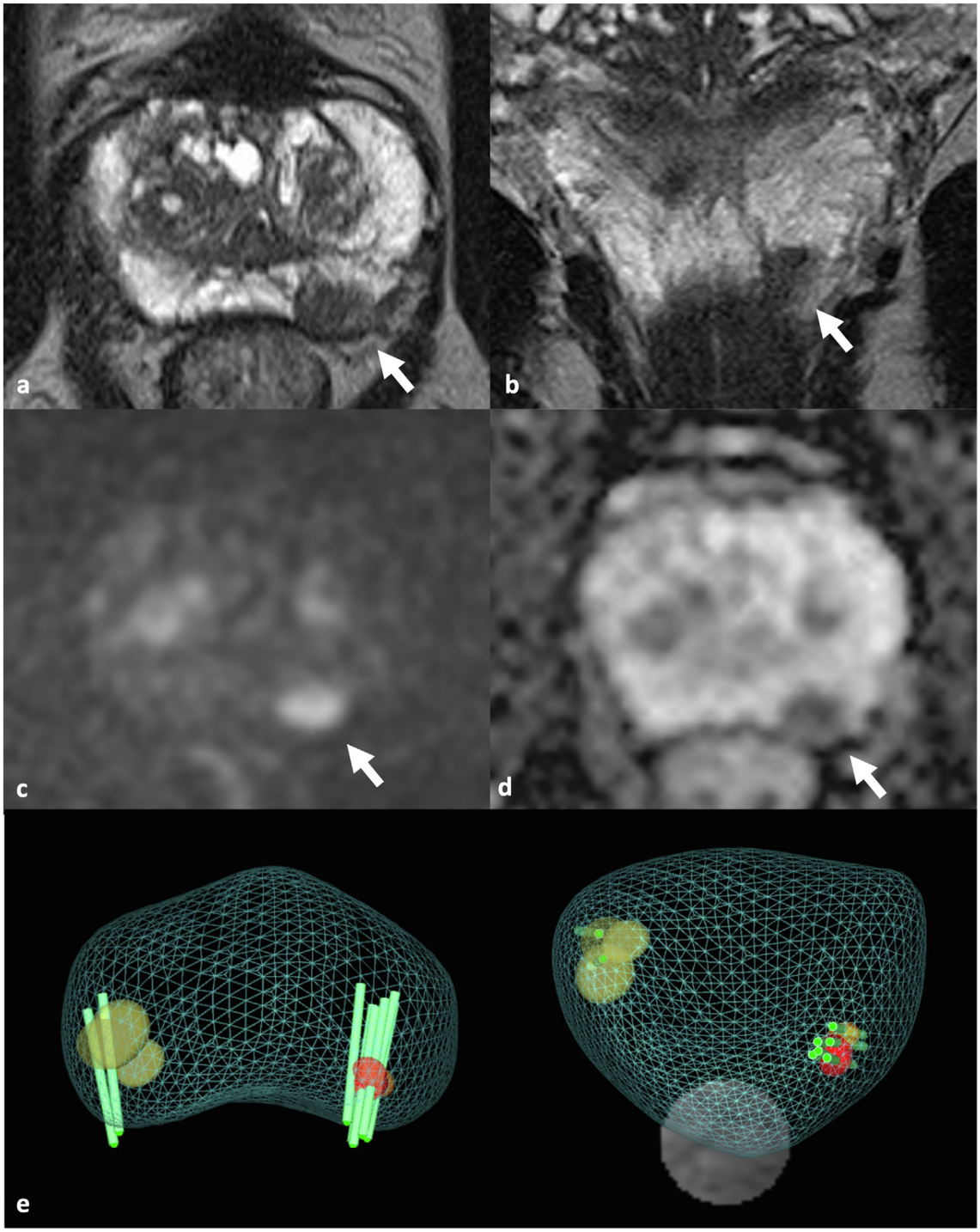


Fig. 2 – Case example: a 59-yr-old man with normal PSA levels (2.78 ng/ml) and no family history of prostate cancer was randomized into arm A and subsequently underwent biparametric MRI (bpMRI), regardless his PSA values. As indicated by the white arrows, imaging revealed a focal hypointense lesion in the left posterolateral peripheral zone on T2-weighted (A) axial and (B) coronal images. This lesion also demonstrated focal hyperintensity on (C) high b-value diffusion-weighted imaging and (D) corresponding focal hypointensity on the ADC map. Based on these findings, the lesion was classified as PI-RADS 4. (E) The patient proceeded to a targeted biopsy using the TRUS-MRI fusion technique, which confirmed a diagnosis of prostate cancer with a Gleason score of 7 (4 + 3). As demonstrated in Fig. 2E, a contralateral alteration (which was classified as PI-RADS 3, and therefore not included in the final analysis) was detected and sampled. ADC = apparent diffusion coefficient; MRI = magnetic resonance imaging; PI-RADS = prostate imaging reporting and data system; PSA = prostate-specific antigen; TRUS = transrectal ultrasound.

improved ICER of €1635.71 per additional GG ≥2 cancer case detected, representing a 25.7% (95% CI 20.7–30.8%) improvement compared with the base-case ICER of €2201.75.

3.6. Safety and adverse events

No serious adverse events or procedure-related deaths occurred. Nonserious adverse events were limited to mild,

Table 3 – Overall costs associated with arms A and B, evaluated through a health technology assessment approach

	Patients' history collection	Laboratory testing		Diagnostic imaging	Prostate biopsy		Total
		Acceptance and blood draw	Analysis of blood sample		Procedure	Analysis of the sample	
Arm A							
Instrument costs							
Total costs per patient	–	€0.55	€6.17	€32.17	€410.66	€0.50	€450.05
Total consumed costs	–	€202.95	€2276.73	€11 870.73	€16 426.40	€20.00	€30 796.81
Staff costs							
Total costs per patient	€0.83	€1.33	€1.35	€21.08	€49.49	€11.05	€85.13
Total consumed costs	€306.27	€490.77	€498.15	€7778.52	€1979.60	€442.00	€11 495.31
Total costs							
Total costs per patient	€0.83	€1.88	€7.52	€53.25	€460.15	€11.55	€535.18
Total consumed costs	€640.54	€4324.06	€2774.88	€23 409.82	€19 311.89	€462.00	€42 292.12
Arm B							
Instrument costs							
Total costs per patient	–	€0.55	€6.17	€32.17	€410.66	€0.50	€450.05
Total consumed costs	–	€209.10	€2362.00	€4182.35	€8213.16	€10.07	€14 979.06
Staff costs							
Total costs per patient	€0.83	€1.33	€1.35	€21.08	€49.49	€11.05	€85.13
Total consumed costs	€319.68	€510.34	€518.32	€2740.96	€989.70	€221.00	€5295.53
Total costs							
Total costs per patient	€0.83	€1.88	€7.52	€53.25	€460.15	€11.55	€535.18
Total consumed costs	€664.85	€4488.12	€2880.16	€8247.36	€9655.94	€231.00	€20 274.59

self-limited postbiopsy symptoms, including low-grade fever (3.3%), transient hematuria, or rectal discomfort. All cases resolved spontaneously or with simple symptomatic treatment (antipyretics and short antibiotic therapy). No hospitalizations or significant complications were recorded. No MRI-related adverse effects occurred, as all participants had been prescreened for contraindications.

4. Discussion

The randomized PROSA trial demonstrated that bpMRI can be integrated effectively into PCa screening. The MRI-first strategy (arm A) identified a 4.6% rate of csPCa compared with 1.8% in the PSA-based strategy (arm B), corresponding to an RD of 2.8 pp (95% CI 0.3–5.4) and an RR of 2.5 (95% CI 1.1–6.1). The overall PCa detection rate (ciPCa + csPCa) was also higher in arm A. Importantly, this increase in detection concerned clinically significant disease, as the rates of low-grade PCa (GG1) and negative/inflammatory findings were similar between groups (RD 1.4 pp, 95% CI −0.3 to 3.1; RR 2.3, 95% CI 0.7–7.5; and RD 1.5 pp, 95% CI −0.6 to 3.6; RR 1.7, 95% CI 0.7–3.8, respectively). These results indicate that the MRI-first approach improved the detection of GG $\geq$2 PCa, without clear evidence of an increased potential risk of overdiagnosis, as already proved in patients with a clinical suspicion [9].

Nearly half of men diagnosed with GG $\geq$2 cancer in arm A had normal PSA levels (47.1%), underscoring the limitations of PSA as a first-line test and the sole screening tool. These results are in line with those already present in the literature [28,30,31], and these support a broader integration of contrast-free MRI into the PCa screening algorithms.

The median PSA levels were within the normal range in both arm A (2.26 ng/ml) and arm B (2.00 ng/ml), although these were slightly higher than those reported in previous screening studies [18,32]; moreover, the number of men with increased PSA levels were higher than the number expected from the sample size calculation. These differences may reflect a partially self-selected population, as participants were enrolled from our institution, representing individuals with greater awareness or concern for prostate health, and possibly more inclined to volunteer for participation. Although in arm A all men underwent bpMRI irrespective of PSA, the PSA distribution of our cohort (median $\approx$ 2.0 ng/ml, with a relatively smaller proportion of men with PSA <1.0 ng/ml) indicates a higher baseline risk than would be expected in an unselected population-based screening cohort. This difference in the underlying risk may influence the observed MRI yield and the number of MRI scans needed per GG $\geq$2 cancer case.

The benefit-harm ratios help contextualize pathway performance [33]. Grade selectivity captures preferential detection of GG $\geq$2 PCa over GG1 disease; this ratio was slightly higher with the MRI-first strategy, indicating more selective detection. Biopsy efficiency reflects the GG $\geq$2 yield relative to unproductive biopsies (GG1 + benign) and similarly favored the MRI-first strategy. Biopsy avoidance quantifies selective avoidance of unnecessary procedures and was substantially higher in arm A, consistent with the broader avoidance in a first-line MRI pathway. While differences in selectivity and efficiency were modest, the avoidance ratio suggests a more favorable balance of benefits and harms with the MRI-first strategy, noting that arm B estimates pertain to PSA-positive men who underwent MRI. Moreover, the resulting numbers needed to biopsy are comparable with the meta-analytic estimates for MRI-based screening pathways (approximately two biopsies per case of GG $\geq$2 cancer) [34], supporting that the integration of both MRI strategies in screening pathways (both MRI-first and PSA-triggered MRI strategies) achieved high biopsy efficiency in line with the broader literature [33].

The NNT, also referred to as the number needed to screen (NNS) in this context [35], indicates that in our trial, 36 prostate bpMRI scans were required to detect one addi-

tional case of GG ≥2 PCa. Although the difference in csPCa detection between the arms was at the threshold of statistical significance ($p = 0.05$), its clinical importance should also be considered. The absolute increase of 2.8 pp and the corresponding NNS of 36 indicate that one additional case of csPCa was detected for every 36 men screened with bpMRI instead of PSA testing. This modest absolute effect nonetheless represents a clinically meaningful improvement in early detection, particularly as it was achieved without increasing overdiagnosis or biopsy-related harms. Thus, while the statistical evidence is borderline, the clinical impact is relevant in the screening context.

We implemented a cost-effective analysis, which aims to compare interventions by calculating costs per unit of health outcome [36,37], which yielded an ICER of €2201.75 per additional case of GG ≥2 cancer detected. Moreover, the MRI-first approach maintained its cost effectiveness across wide (±50%) variations in costs and detection rates, underscoring the robustness and generalizability of the economic results. Although this ICER provides a meaningful measure of efficiency, it cannot be compared directly with conventional willingness-to-pay thresholds, which are typically expressed in cost per QALY gained. Our intent was to implement a cost-effective analysis, adopting a health care payer perspective and a single-round horizon, capturing direct medical costs but not downstream utilities/costs of treatment, progression, complications, or overdiagnosis. We could not implement a cost-utility analysis, which is a specialized form of a cost-effective analysis that incorporates outcomes in terms of QALYs. Given the 1-yr follow-up period of our study, it is not possible to accurately capture the evolution of key health states that determine QALYs in PCa, such as transitions between active surveillance, definitive treatment, metastatic progression, or terminal disease [29]. To contextualize and bridge our findings toward broader HTA assessments, we recalculated the ICUR using a QALY-per-cancer estimate [29]. With an ICUR of €8468 per QALY gained, MRI-first screening is positioned as a cost-effective strategy relative to PSA-first screening, particularly in health care settings where thresholds range from €20 000 to €50 000 per QALY. Although a QALY-per-cancer approach simplifies the full spectrum of clinical heterogeneity, it allows for practical modeling where individual-level utility data are not available. Indeed, recent QALY-based evaluations of MRI-enhanced screening report ICURs generally within accepted the willingness-to-pay ranges: €11 355 per QALY for triennial PSA with reflex mpMRI and MRI-guided biopsy (health care payer perspective) [38], US$53 736 per QALY for PSA + MRI with combined biopsies in the STHLM3-MRI modeling (lifetime horizon) [39], and US$6000–21 131 per QALY for MRI-guided biopsy strategy (10-yr horizon) [40]. These values lie near commonly cited European willingness-to-pay bands (≈€20 000–50 000 per QALY) used to classify costs per QALY [39,41]. Although actual QALYs were not assessed in this study, the low ICER per additional csPCa diagnosis suggests a potentially favorable economic profile, in line with the thresholds commonly used in QALY-based evaluations. Extension of the present model to a longer time horizon that includes active surveillance, treatment, and metastatic disease health states would allow future research to translate real health states into concrete QALY-based outcomes.

Indeed, from a health system's perspective, early diagnosis through MRI is associated with significant downstream savings. Advanced-stage PCa management can cost up to €140 000 annually per patient, while early detection reduces this burden by €125 000–130 000 per patient [42–44]. Although downstream costs such as overtreatment, patient anxiety, or productivity loss were not included in this analysis, and these can be consistent for GG ≥2 PCa, the presented results help support the cost effectiveness of an early intervention.

MRI-first screening with cost-effective bpMRI protocols appears to be a clinically and economically advantageous alternative to PSA-based strategies, improving csPCa detection and showing favorable ICER and NNS, linked to a higher biopsy rate but without a corresponding increase in low-grade cancer diagnoses. Longer follow-up and QALY-based analyses are needed to inform broader HTAs.

Although our findings suggest that contrast-free MRI screening can be economically sustainable on a per-patient basis, broader implementation would require careful consideration of system-level constraints, including imaging capacity, radiologist availability, and existing diagnostic backlogs, which may limit scalability in certain health care settings.

A PI-RADS 3 threshold was adopted to enhance cancer detection in a younger screening population, acknowledging anatomical differences in younger men [45,46] and the uncertain predictive value of PI-RADS 3 [47,48]. Indeed, most csPCa cases were linked to PI-RADS 4, while ciPCa and benign findings were predominantly associated with PI-RADS 3. In our exploratory analysis, raising the biopsy threshold from PI-RADS ≥3 to PI-RADS ≥4 improved both the clinical and the economic performance of the MRI-first pathway. This approach would avoid 26 of 60 MRDB procedures (43.3%), predominantly from the MRI arm, while deferring only two csPCa cases (8.3%, one per arm). Such a reduction improves the balance between diagnostic benefit and procedural harm substantially, minimizing unnecessary biopsies and the detection of indolent disease, while maintaining a high yield for clinically significant cancers. The avoided procedures would generate approximately €12 264.20 in payer savings, lowering the ICER from €2201.75 to €1635.71 per additional GG ≥2 cancer case detected (–25.7%). Overall, these findings indicate that adoption of a PI-RADS ≥4 threshold could enhance diagnostic efficiency and cost effectiveness, strengthening the value proposition of MRI-based PCa screening. As a minority of csPCa resides in PI-RADS 3, a risk-adapted triage of PI-RADS 3 (eg, incorporating PSA density or family history) can help preserve sensitivity while retaining most of the avoidance benefits [14,49].

This study has several limitations that should be acknowledged. The relatively small number of csPCa cases reduced the statistical power to detect modest between-group differences. Moreover, the recruitment of a self-selected employee cohort from a single institution may have introduced a selection bias, as participants likely had

higher health awareness and better access to follow-up care than the general population. This could have led to a slight overestimation of adherence and diagnostic performance compared with population-based screening programs, and may partly explain the relatively higher PSA values observed in our cohort. Indeed, the PSA distribution in our cohort does not fully reflect that of an unselected screening population, which should be characterized by a higher proportion of men with PSA values <1.0 ng/ml. This difference in baseline risk distribution may have influenced the diagnostic yield observed in our study. As such, the generalizability may be limited, and future evaluations in cohorts with a PSA distribution representative of the general screening population are warranted. A confirmatory PSA test was not performed prior to MRI in arm B, which may have led to some unnecessary MRI scans due to transient PSA elevations. A lack of participant blinding, although unavoidable given the nature of the interventions, might have influenced downstream clinical decisions, such as biopsy acceptance, introducing a degree of performance bias. In addition, a small proportion of MRI examinations were excluded due to suboptimal image quality, which may have caused a minor selection bias by preferentially retaining higher-quality scans. The exclusive use of MRI-targeted biopsies may have slightly underestimated the prevalence of indolent or multifocal low-grade disease; however, this aligns with the objective of prioritizing the detection of clinically significant disease. Finally, the short follow-up period and single-round screening design precluded evaluation of long-term outcomes such as mortality reduction or overdiagnosis, potentially limiting inferences about the population-level impact of MRI-based screening. These methodological choices were intentional, allowing an early, pragmatic evaluation of diagnostic performance and cost effectiveness in a real-world screening setting. Moreover, the absence of long follow-up prevents from evaluating crucial health states that are necessary to comprehensively implement a cost-utility analysis. To overcome this limitation, we implemented a further analysis using a fixed estimate of QALYs per cancer detected. While evidence based, this simplification does not account for differences in tumor aggressiveness, patient age, comorbidities, or treatment-related quality-of-life adjustments. Additionally, this approach potentially underestimates harms from overdiagnosis or overtreatment, which would reduce the net QALY gains in lower-risk cases. Nonetheless, concrete QALYs were not calculated in this study, as the analysis was designed as a cost-effectiveness assessment per diagnosis rather than a cost-utility analysis based on QALYs. The ICER remained favorable, reflecting the low marginal cost of contrast-free MRI and the diagnostic advantage observed in arm A, supporting the feasibility of MRI-based screening. Moreover, cost estimates were derived from Italian reimbursement rates, which may limit the generalizability of absolute cost figures to other health care systems. However, the use of ICERs mitigates this limitation by emphasizing the relative differences in costs and outcomes, which are more transferable across settings.

Overall, these factors may modestly affect precision or generalizability, but are unlikely to change the main conclusion that an MRI-first screening strategy improves the detection of csPCa without increasing the risk of overdiagnosis.

5. Conclusions

In this randomized screening trial, a contrast-free MRI-first pathway improved the detection of csPCa versus PSA-triggered MRI, maintained a comparable detection rate of GG1 disease, and showed a favorable benefit-harm profile. At the population level, the MRI-first approach achieved a low NNS to detect one additional case of GG $\geq$2 cancer and a favorable ICER and ICUR. These findings provide pragmatic, population-level evidence that bpMRI is a viable screening option, particularly given that a substantial proportion of GG $\geq$2 cancer cases occurred at normal PSA levels, underscoring the added value of MRI in risk-adapted strategies.

***Author contributions*:** Valeria Panebianco had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study concept and design: Panebianco, La Torre, Messina.
Acquisition of data: Borrelli, Laschena, Antonini, Shaholli, Magliocca, Sciarra.
Analysis and interpretation of data: Messina, Borrelli, Novelli, Laschena, Antonini, Santini, Magliocca.
Drafting of the manuscript: Messina, Borrelli, Antonini.
Critical revision of the manuscript for important intellectual content: Panebianco, Novelli, La Torre, Sciarra, Santini.
Statistical analysis: Messina, Borrelli, Novelli, La Torre, Antonini, Shaholli.
Obtaining funding: Panebianco.
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Data sharing statement: Deidentified individual participant data, along with the data dictionary and study protocol, will be made available upon

reasonable request, beginning with publication. Data access will be granted for academic, noncommercial use following approval of a research proposal and signing of a data access agreement. Requests should be submitted to the corresponding author's e-mail address: valeria.panebianco@uniroma1.it.

Supplementary material

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